

- Q.28 Discuss Piston operated valve & diaphragm operated valve.
- Q.29 Write the frequency range of Electromagnetic spectrum VLF, LF, MF, HF, VHF, UHF, Microwave.
- Q.30 Discuss the different types of modulation systems.
- Q.31 Explain the Block diagram of AM receiver.
- Q.32 Explain the SSB system and draw its spectrum.
- Q.33 Basic block diagram and principle of working of the following FSK, PSK.
- Q.34 Discuss the Frequency Hopping Technique.
- Q.35 Explain the Analog and digital modulation.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
- Q.36 Compare and contrast the performance of open-loop and closed loop control systems in terms of accuracy, stability, disturbance rejection and complexity.
- Q.37 Derive the mathematical expression for a PID (Proportional-Integral-Derivative) controller, Discuss how PID controllers improve system performance in terms of stability, transient response, and steady-state error.
- Q.38 Discuss the following terms.
- Vestigial side band systems.
 - Piston operated valve
 - DSB
 - QPSK

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5th Sem / Mechatronics Sub : Process Control & Data Communication

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The transfer function is defined as
- a) Output/Input
 - b) Input/Output
 - c) Output x Input
 - d) Output + Input
- Q.2 Time invariance means the system properties:
- a) Depend on time
 - b) Change with frequency
 - c) Are constant over time
 - d) Are nonlinear over time
- Q.3 A disadvantage of a closed-loop system is :
- a) Less accuracy
 - b) Complexity and higher cost
 - c) Low sensitivity to disturbances
 - d) No feedback path
- Q.4 Which of the following is used to measure pressure
- a) RTD
 - b) Thermocouple
 - c) Bourdon tube
 - d) Ultrasonic sensor
- Q.5 Which device converts temperature into electrical signals
- a) Bourdon tube
 - b) Thermocouple
 - c) Orifice plate
 - d) Ultrasonic sensor

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- Q.6 Which controller is best for eliminating offset in steady state
- P
 - PI
 - PD
 - P-only
- Q.7 A PI controller includes which two actions
- Proportional and derivative
 - Proportional and integral
 - Derivative and Integral
 - None of the above
- Q.8 What type of energy input is required to operate a solenoid valve.
- Hydraulic
 - Pneumatic
 - Electrical
 - Mechanical
- Q.9 In Frequency Modulation (FM), the information is carried by.
- Amplitude of carrier
 - Frequency of carrier
 - Phase of carrier
 - Both A & B
- Q.10 ASK stands for
- Amplitude standard Keying
 - Amplitude Shift Keying
 - Analog Shift Keying
 - Asynchronous Signal Keying

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 FSK stand for _____.
- Q.12 Which type of system has no feedback loop.

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- Q.13 What instrument is commonly used to measure fluid flow.
- Q.14 Which part of the PID controller reacts to the rate of change of error.
- Q.15 Write one disadvantage of an On-Off controller.
- Q.16 Control valve is a final control element in a control system (Yes/No)
- Q.17 What are the two types of sidebands in AM called.
- Q.18 What is the phase difference between adjacent QPSK symbols.
- Q.19 What type of demodulator is used in FM receivers.
- Q.20 DSB-SC stand for _____.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Define feedback in control system. What are the advantages of using feedback in control systems.
- Q.22 Explain the component of closed loop control system with a block diagram.
- Q.23 What is a manipulated variable (MV)? How is it used in control.
- Q.24 Define a PID controller and explain the role of each term.
- Q.25 Draw the graph of ($t > 0$ & $t \leq 0$) for step and ramp test inputs.
- Q.26 Explain the Principle of operation and constructional details of solenoid valves.
- Q.27 What are the different types of characteristics of control valves? Explain each.

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